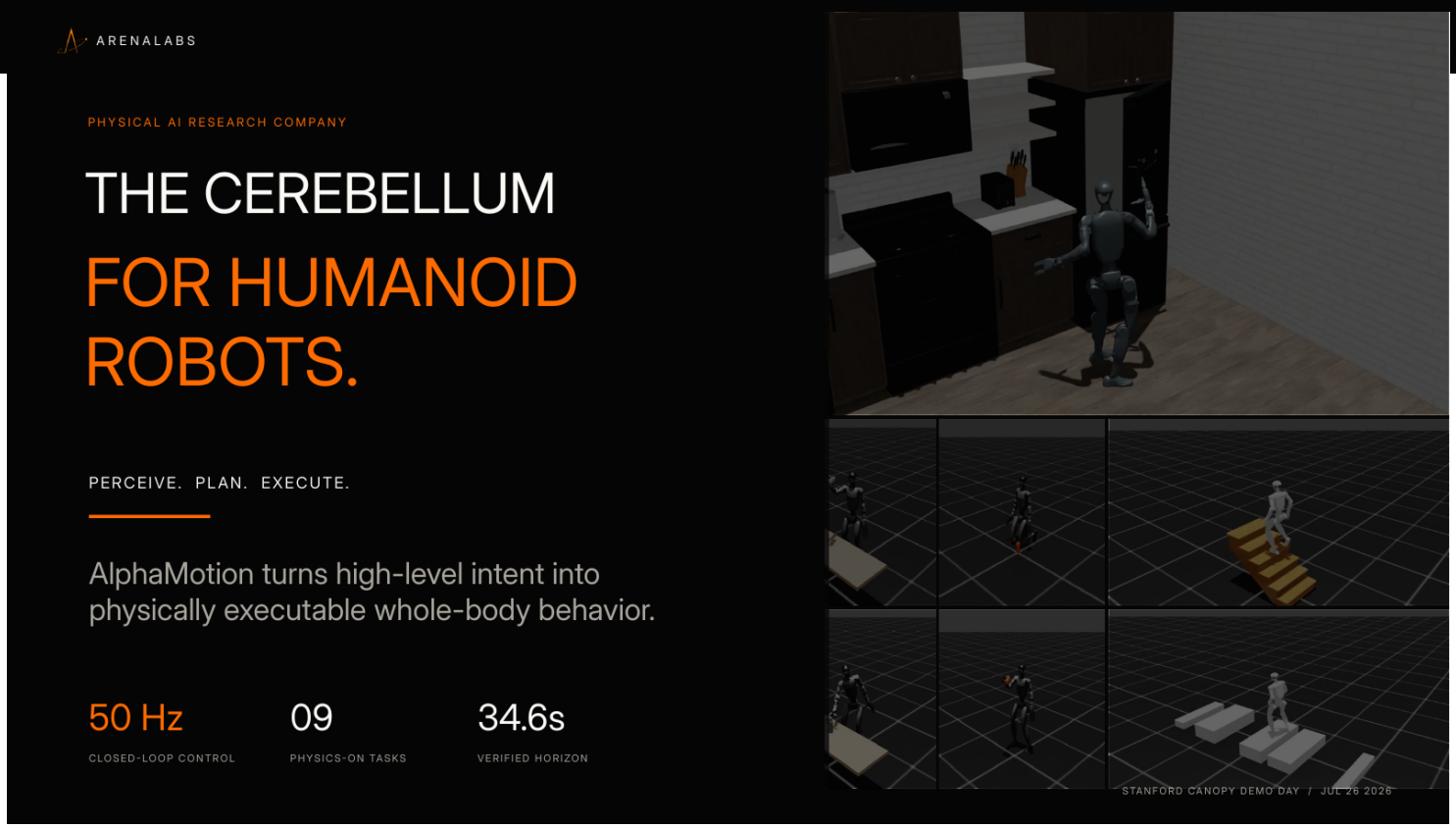
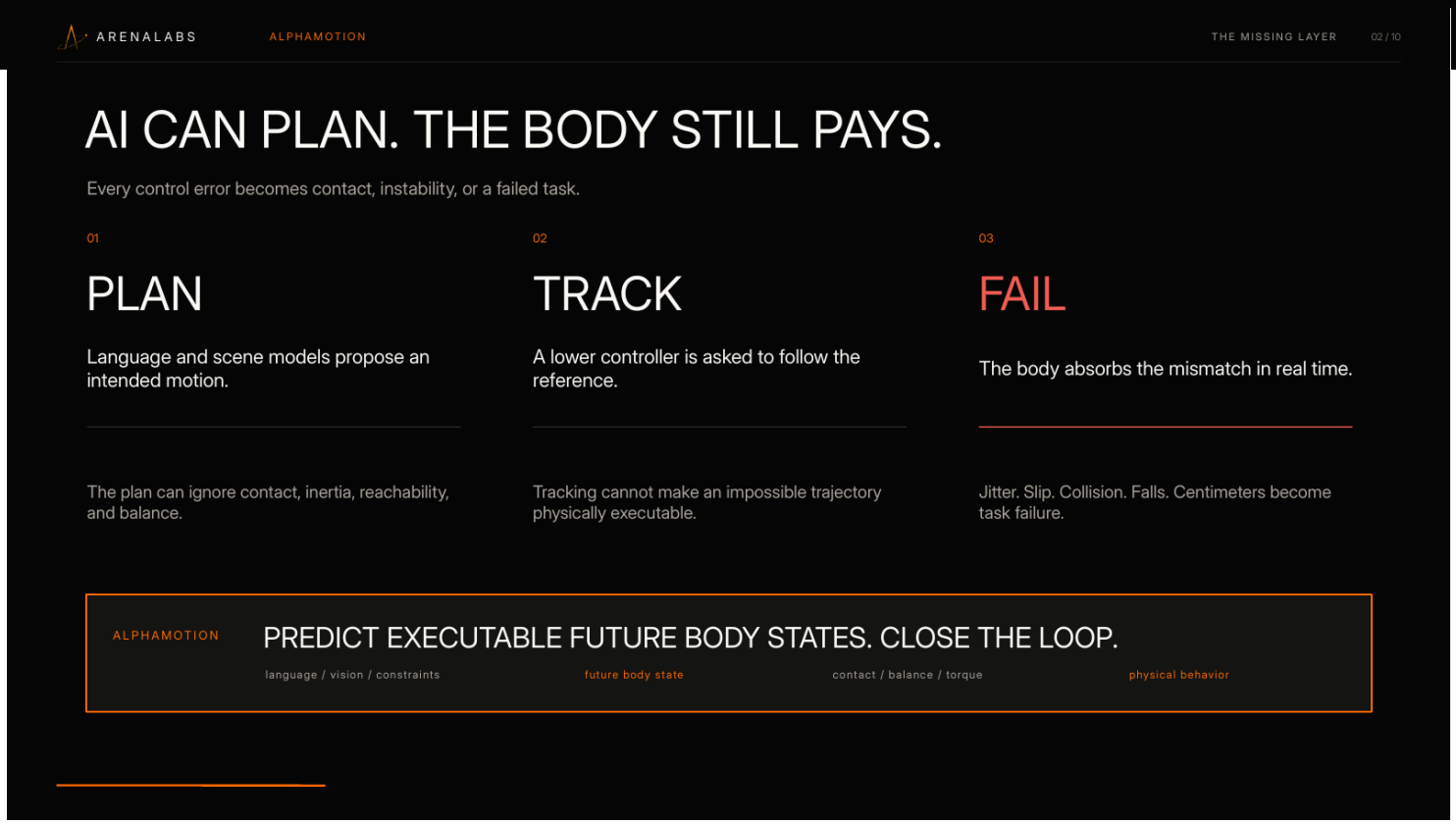


0:00–0:15

Advance at 0:15

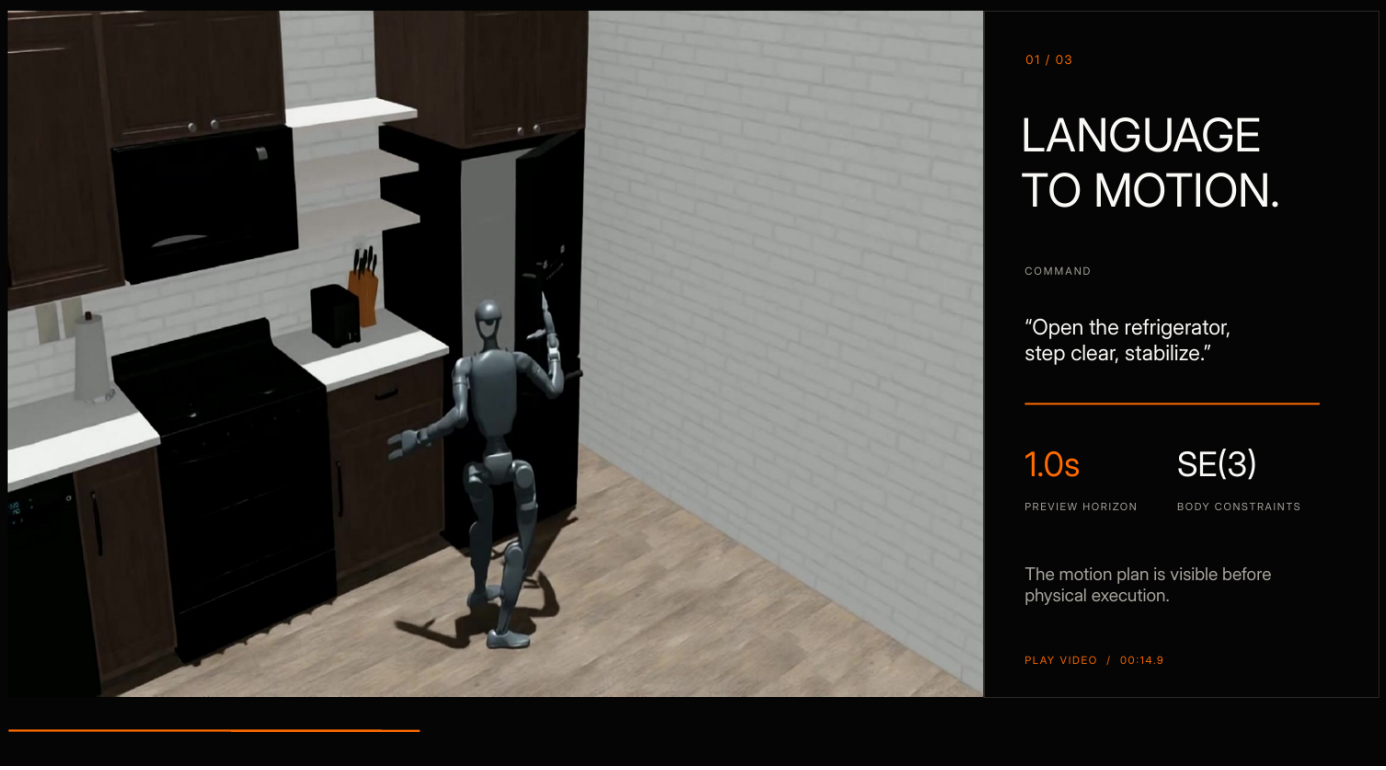
Hi everyone. I'm Lloyd, founder and CEO of ArenaLabs.
We are a Physical AI research company building the cerebellum for humanoid robots.
Our first product is AlphaMotion: whole-body intelligence that turns high-level intent into physically executable behavior.



0:15–0:43

Advance at 0:43

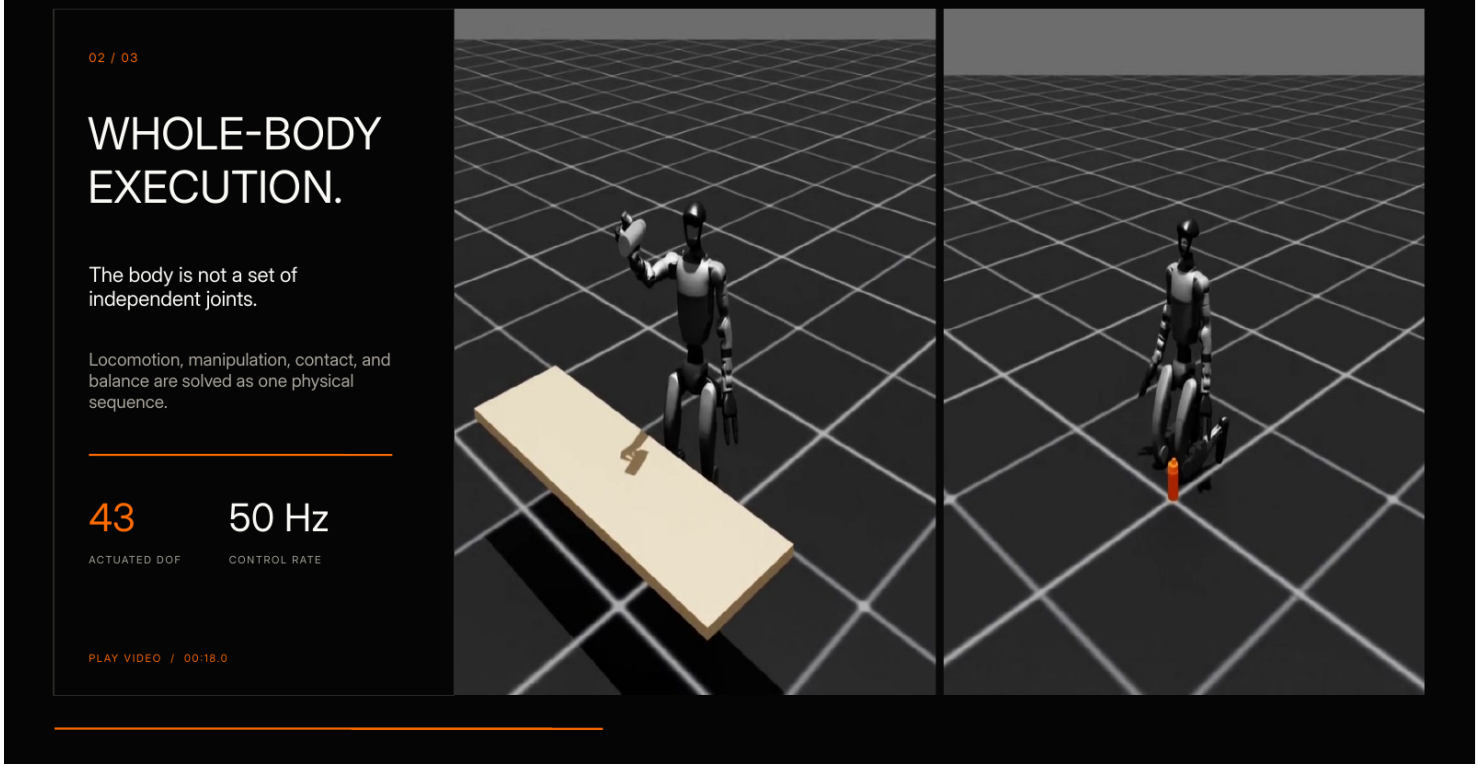
Robots can understand scenes, language, and multi-stage tasks.
But the body still pays for every error.
The planner can propose a motion without understanding contact, inertia, reachability, or balance. A tracking controller is then asked to follow it, even when the reference is physically impossible.
The result is jitter, slip, collision, falls, and manipulation failures caused by only a few centimeters of error.
AlphaMotion predicts executable future body states and closes the loop between intent and the physical body.



0:43–1:02

Play clip; advance when it ends

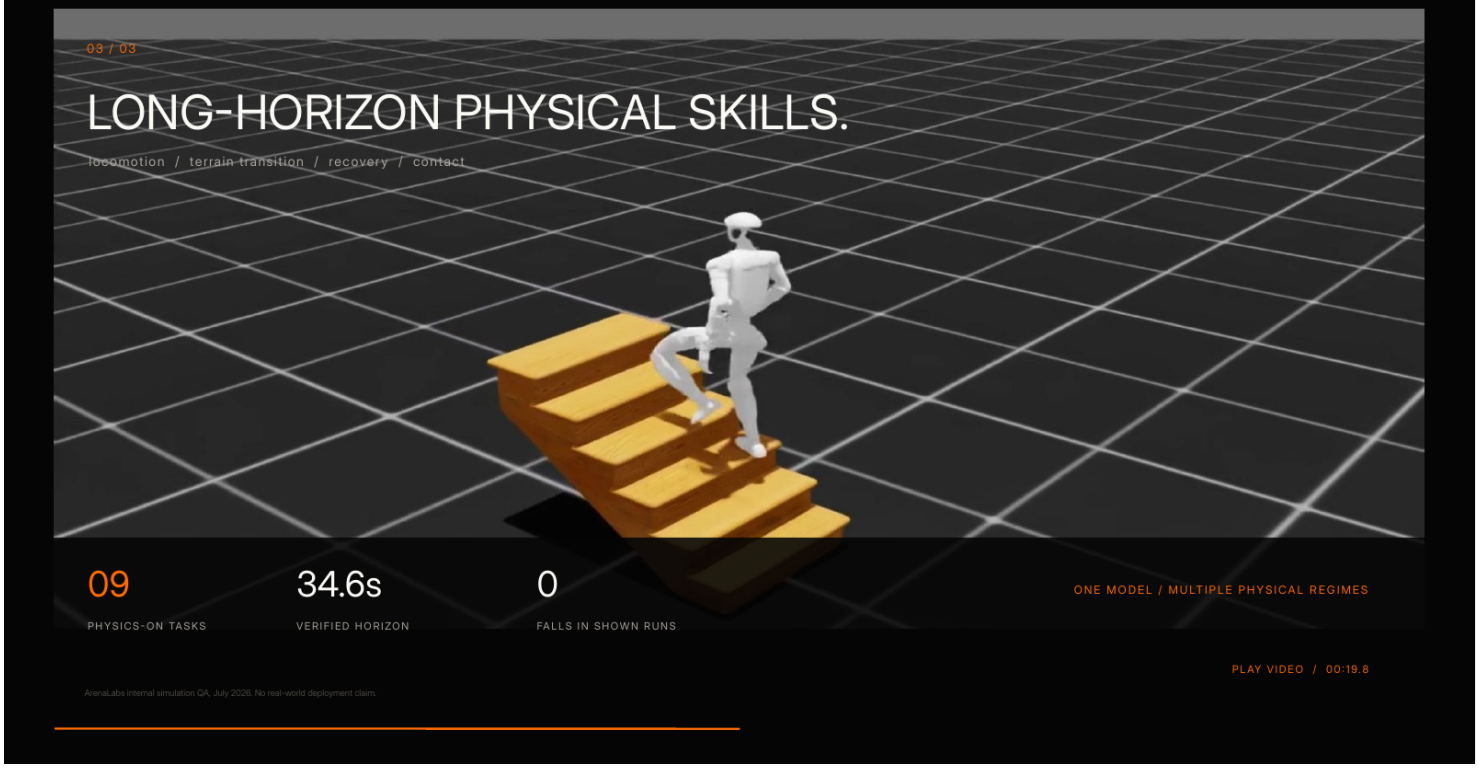
First, a natural-language command becomes a staged whole-body plan.
The future plan is rendered ahead of execution, while the physical body follows it through the control loop.
The important point is not the refrigerator. It is that language is translated into body-level constraints before the robot commits to contact.



1:02–1:25

Play clip; advance when it ends

Second, the model coordinates the whole body.
The upper body cannot reach accurately if the feet, pelvis, and center of mass are unstable.
AlphaMotion treats locomotion, manipulation, contact, and balance as one sequence, and executes it at a 50-hertz closed-loop control rate.



1:25–1:50

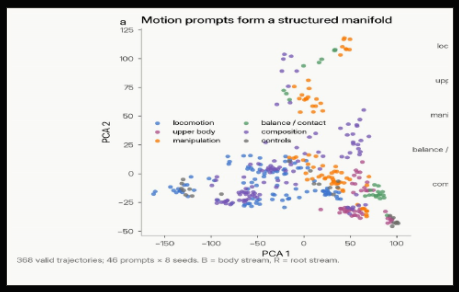
Play clip; advance when it ends

Third, the same behavior layer spans multiple physical regimes: locomotion, terrain transitions, balance recovery, and contact-rich sequences.
These are internal simulation results, not a claim of completed real-world deployment.
What they demonstrate is the architecture's ability to connect a long-horizon plan to a physical control loop.

BEHAVIOR ORGANIZES THE LATENT SPACE.

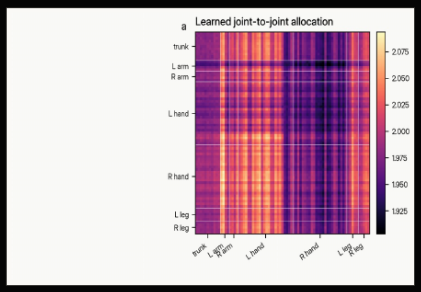
The model separates task structure from sampling noise — and allocates it across joints and time.

PLANNER MANIFOLD



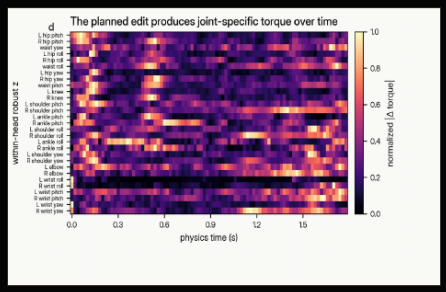
task clusters dominate seed variation

JOINT ALLOCATION



body-part structure emerges in the map

TORQUE × TIME



closed-loop response has temporal structure

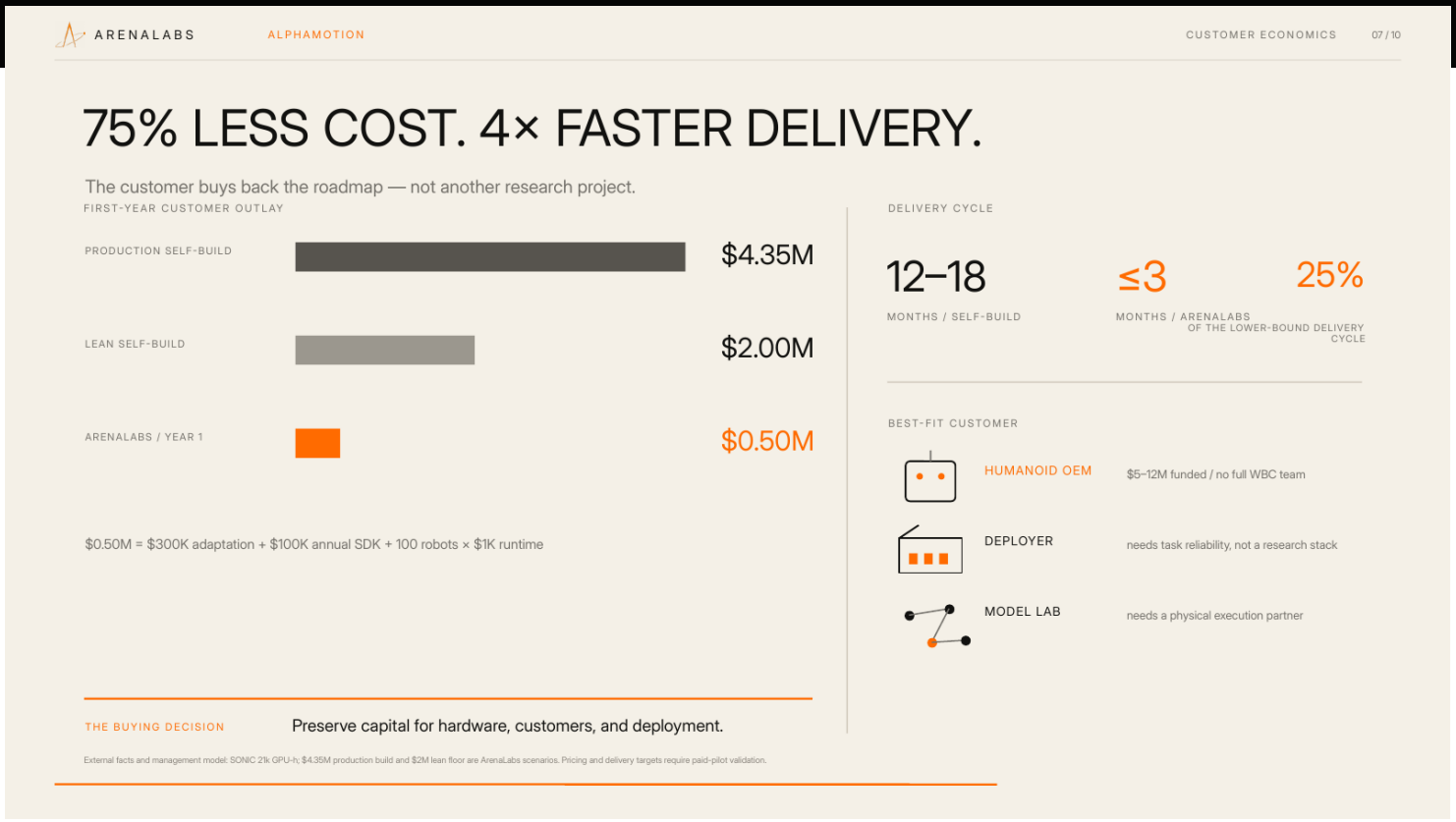
0.826	0.052	368	46 × 8	50 Hz
NMI: PROMPT VS CLUSTER	NMI: SEED VS CLUSTER	VALID TRAJECTORIES	PROMPTS × SEEDS	CLOSED-LOOP READOUT TASK SIGNAL ≫ SEED SIGNAL

Source: ArenaLabs internal representation study, July 2026. PCA/NMI are descriptive analyses of sampled trajectories; causal claims require further intervention tests.

1:50–2:28

Advance at 2:28

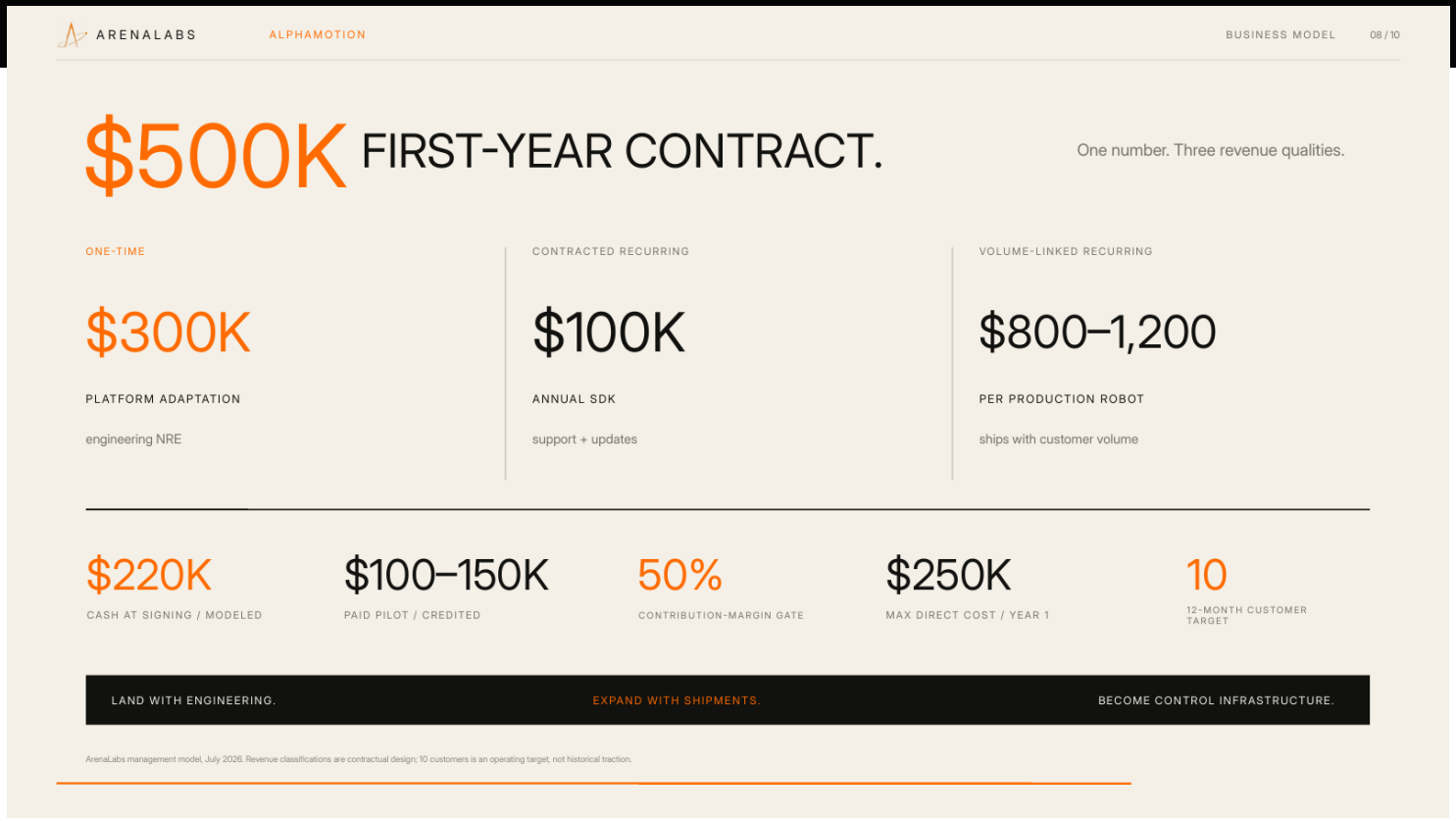
The model is not only producing plausible clips. Its internal representation is structured.
Across 368 valid trajectories, prompt identity explains the clusters far more strongly than random seed: normalized mutual information is 0.826 for task and only 0.052 for seed.
The joint-allocation and torque-time maps show that this structure is distributed across body parts and across the control horizon.
This is internal mechanistic evidence, not a final causal proof. The next experiments intervene on these representations directly.



2:28–3:08

Advance at 3:08

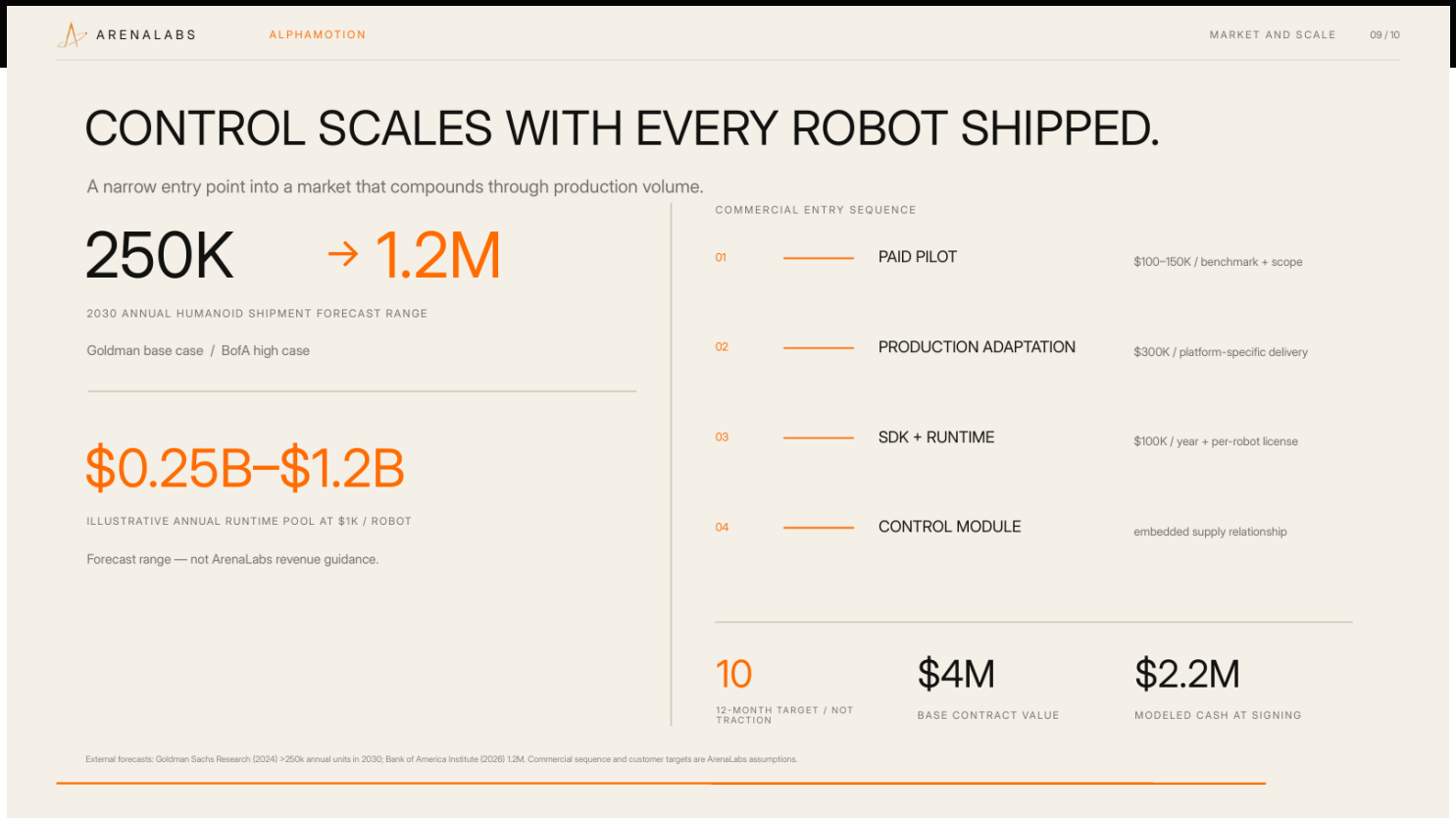
Now the buying decision.
Our target customer is a five-to-twelve-million-dollar-funded humanoid company that cannot spend two to four million dollars and twelve to eighteen months building a control organization from scratch.
At one hundred robots, our modeled first-year outlay is five hundred thousand dollars: seventy-five percent lower than a lean internal build.
Our delivery target is three months or less — one quarter of the lower-bound self-build timeline.
The same platform also serves deployment integrators and foundation-model teams that need reliable physical execution.



3:08–3:42

Advance at 3:42

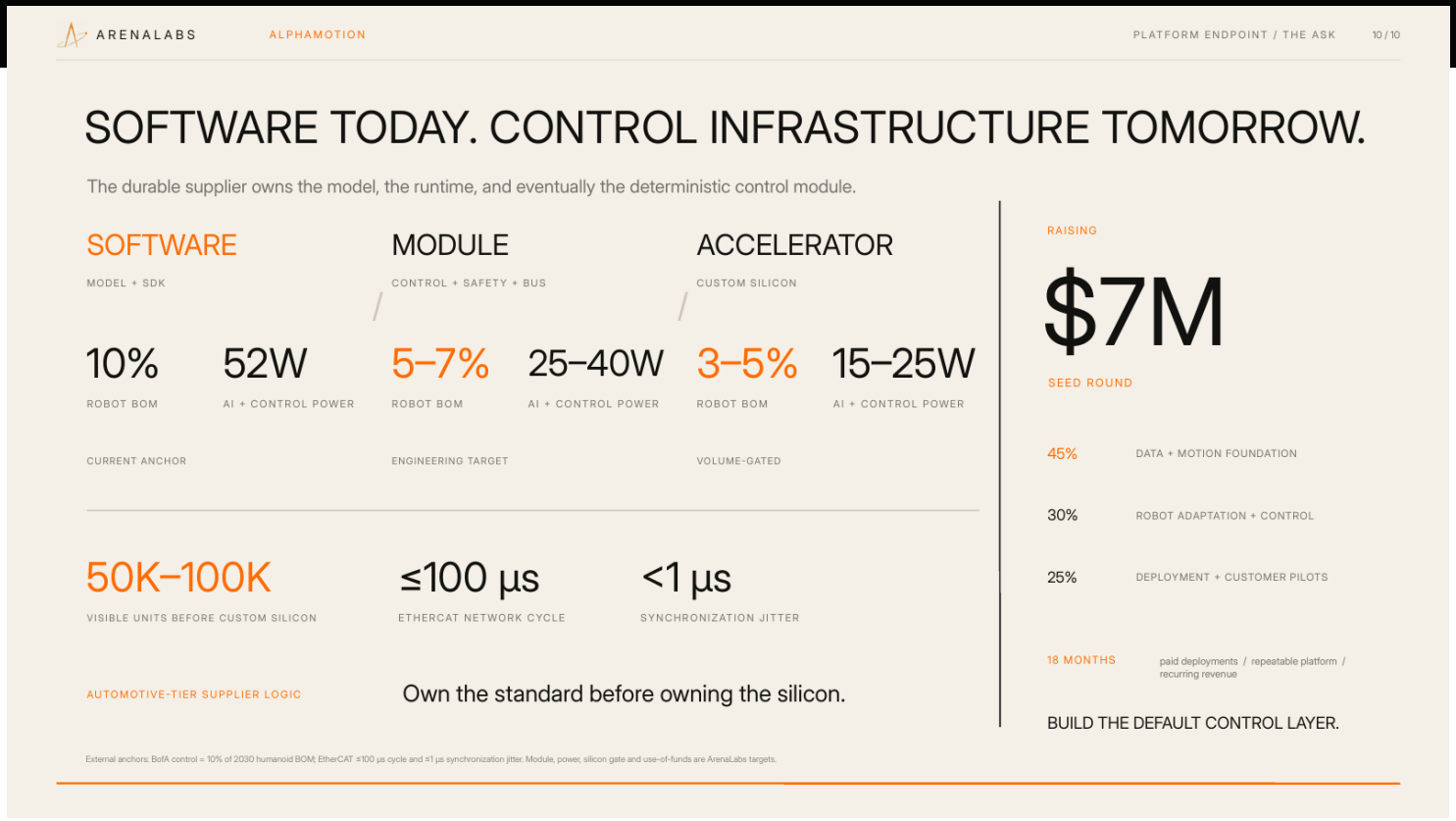
The Airbnb lesson is to make the business model calculable in one line.
Our first-year contract is five hundred thousand dollars: three hundred thousand for platform adaptation, one hundred thousand for the annual SDK, and roughly one hundred thousand from the first one hundred robots.
The quality of revenue improves over time: one-time engineering becomes contracted recurring software and then volume-linked runtime revenue.
We target at least a fifty-percent contribution margin, and enter through a one-hundred-to-one-hundred-fifty-thousand-dollar paid pilot that is credited toward production.



3:42–4:17

Advance at 4:17

The market compounds through every robot shipped.
Published 2030 annual shipment forecasts range from more than two hundred fifty thousand units at Goldman Sachs to 1.2 million at Bank of America.
At one thousand dollars per robot, that implies an illustrative runtime pool of a quarter-billion to 1.2 billion dollars per year. That is market arithmetic, not our revenue guidance.
Our entry sequence is paid pilot, production adaptation, SDK and runtime, and finally an embedded control module.
Our twelve-month operating target is ten customers. It is a target, not traction.



4:17–4:42

Finish at 4:42; keep 18 seconds of buffer

We start with software, but the durable endpoint is control infrastructure.
As the workload stabilizes and production volume becomes visible, the stack can move into an integrated control-and-safety module, and eventually a custom accelerator.
We will not build silicon before the economics justify it. The gate is fifty to one hundred thousand visible committed units.
We are raising seven million dollars to build the motion foundation, adapt it across robot platforms, and convert the first paid pilots into a repeatable deployment platform.
ArenaLabs intends to build the default control layer for Physical AI.